



AW3 Smart Docking

Pallet Docking Solution White Paper

Pallet position, orientation and status for autonomous forklift docking.

Zhejiang MRDVS Technology Co., Ltd.

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Contents

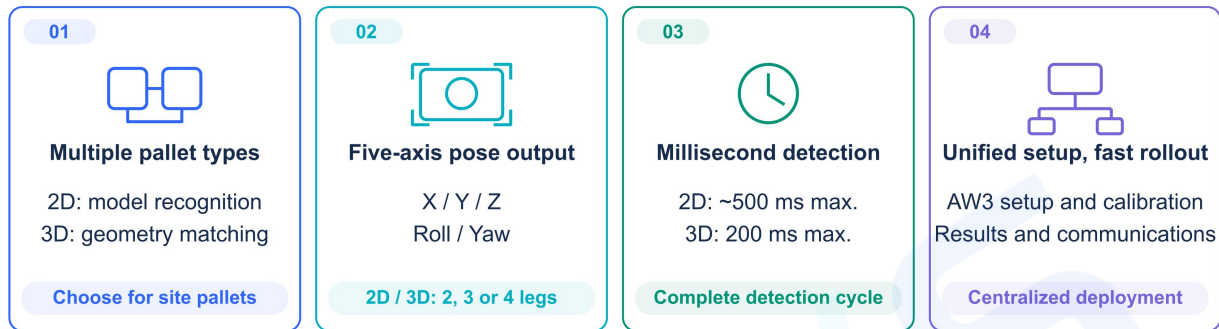
1 Product Overview.....	3
1.1 Key Advantages.....	3
1.2 Application Scenarios.....	3
1.3 Cameras and Computing Platforms.....	4
2 How It Works.....	5
2.1 Two Positioning Methods.....	5
3 Performance and Operating Range.....	6
3.1 Detection Distance and Cycle Time.....	6
3.2 Recognition and Outputs.....	6
3.3 Operating Conditions and Limits.....	7
4 Deployment and Configuration.....	8
4.1 Deployment Workflow.....	8
5 Validation and Acceptance.....	9
6 Frequently Asked Questions.....	10
Revision History and Support.....	11

1 Product Overview

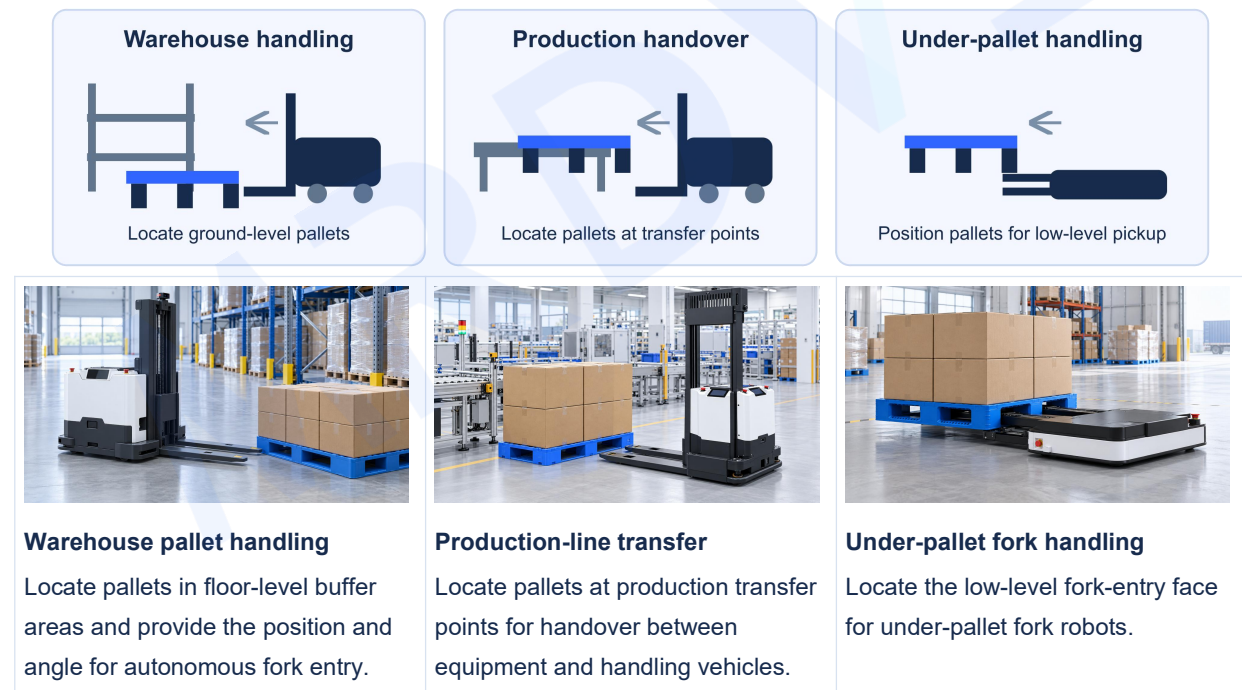
Smart Docking runs on the AW3 Algorithm Platform. It identifies the pallet fork-entry face and supplies the reference point position, orientation and positioning status to autonomous forklifts.

AW3 manages devices, configuration and communication. Pallet Docking provides perception; the vehicle selects targets and controls motion and safety.

1.1 Key Advantages



1.2 Application Scenarios



1.3 Cameras and Computing Platforms

Use S11 with RK3588 for closer-range tasks, or M4 Mega with onboard compute for longer-range tasks. Camera specifications below are not pallet detection limits; see Section 3.1.

Parameter	S11	M4 Mega
Computing platform	External RK3588	Onboard compute, 1 TOPS
Depth technology / illumination	dToF / 940 nm	iToF / 940 nm
Output Data	Depth, RGB, IR amplitude	Depth, RGB, amplitude
Depth Resolution / Frame Rate	240 × 96 Typical 10 / max. 15 fps	640 × 480 Typical 15 / max. 25 fps
RGB Resolution / Frame Rate	1280 × 1080 Typical 10 / max. 15 fps	1280 × 960 Typical 15 / max. 25 fps
TOF Field of View (H×V)	136° × 56° (±3°)	81° × 61° (±3°)
RGB Field of View (H×V)	120° × 110° (±3°)	86° × 55° (±3°)
Depth Measurement Range ¹	Outdoor: 0.1–5 m Indoor: up to 10 m	0.2–5 m
Measurement Accuracy ²	≤ 1 cm	±3 mm + 0.25% × depth
Power Supply	Ethernet: 12–28 VDC USB: 5 V / USB 3.0	24 VDC / 2 A
Average Power Consumption ³	< 4 W	6.8 W @ 24 V
Communication Interfaces	Ethernet / USB	Gigabit Ethernet / CAN / 4 IO
Dimensions	90 × 25 × 25 mm	92 × 47 × 51 mm
IP Rating	Ethernet version: IP54	IP67
Operating Temperature	–20–60°C	–20–60°C

¹ Specification conditions: target reflectivity 10–90%. ² Camera measurement accuracy is not pallet positioning accuracy. ³ S11 power excludes the external RK3588.

2 How It Works

2.1 Two Positioning Methods

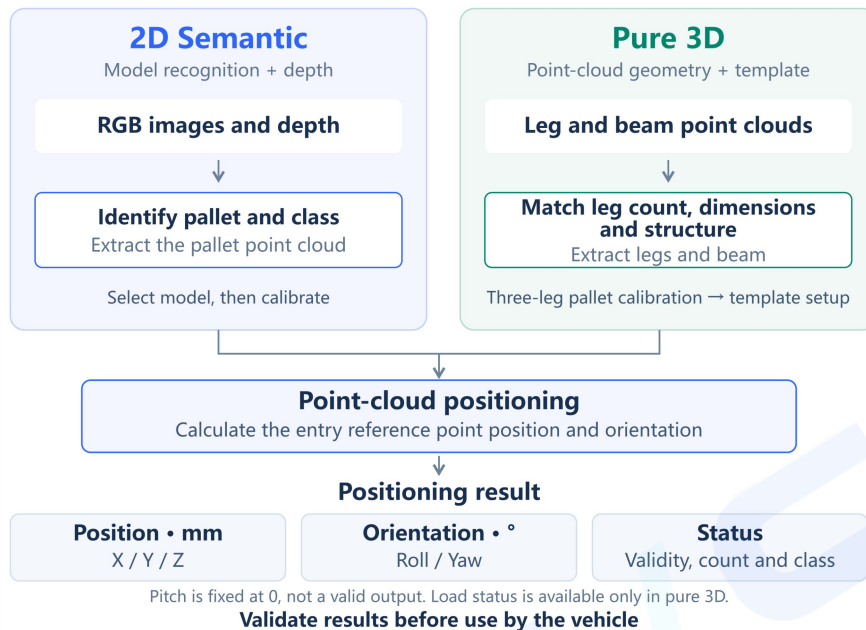
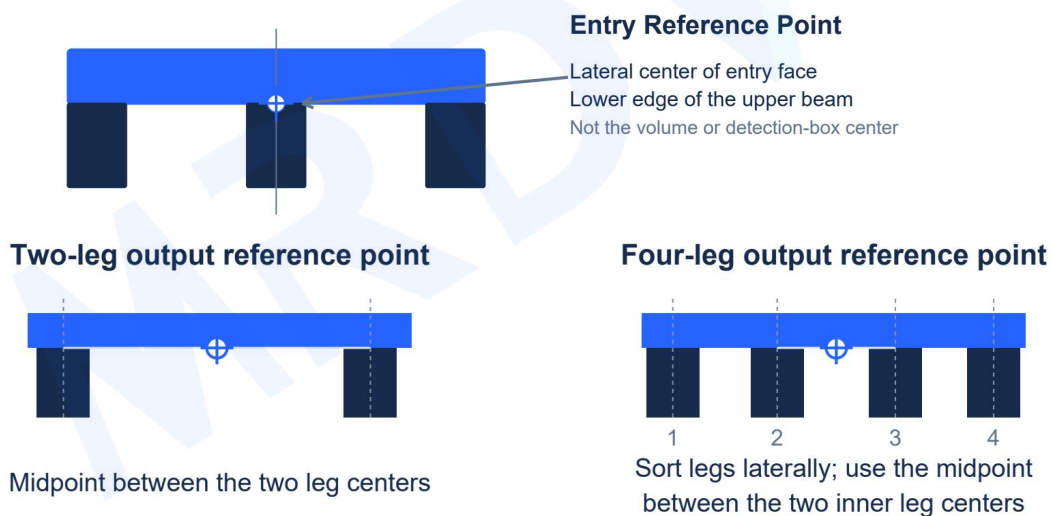


Figure 2 Two recognition methods, point-cloud-based positioning

Entry Face Front View



Benefits and setup trade-offs

2D Semantic Positioning

Benefits

Train models for different site pallet types.
Classify pallets with or without a lower beam.

Preparation and maintenance

Collect site-pallet RGB data for semantic training.
Validate new types; retrain if needed.

Pure 3D Positioning

Benefits

No model training required.
Similar pallet geometry can share a template.

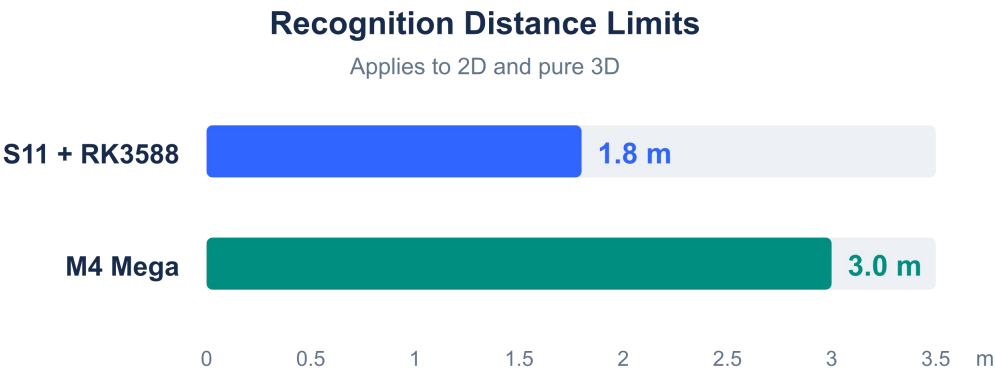
Preparation and maintenance

Tune parameters to site-pallet dimensions.
Many distinct types may need frequent template changes or parameter adjustments.

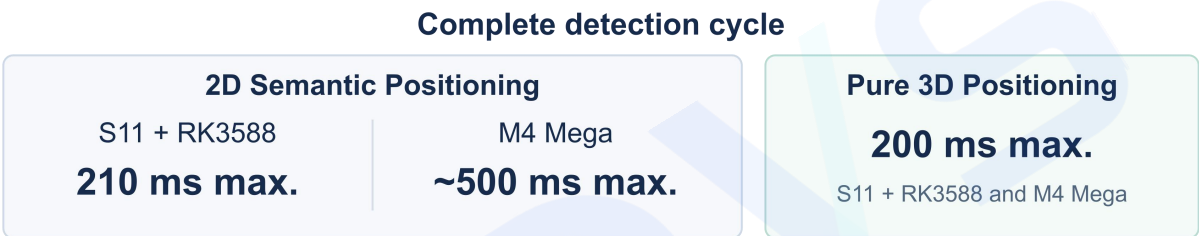
3D is an alternative method, not an automatic failover when 2D fails. Calibrate, configure and validate it before use.

3 Performance and Operating Range

3.1 Detection Distance and Cycle Time

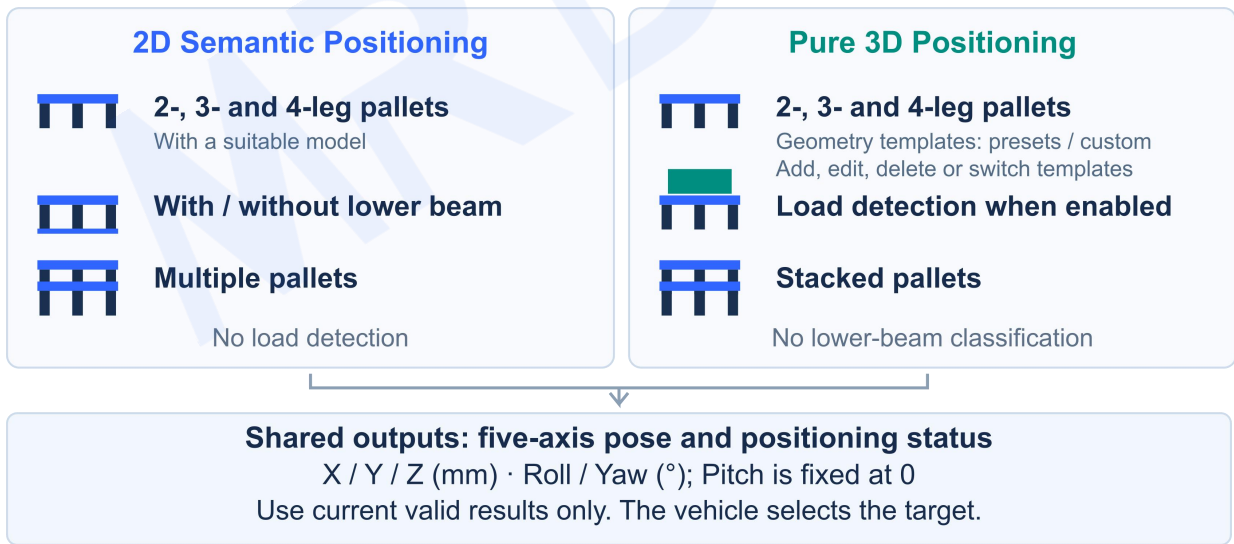


The same distance limits apply to 2D and pure 3D. Validate the usable range with actual site pallets, materials and models.
The minimum recognition distance depends on pallet size and type.



Plan by mode and platform. Validate on site and allow margin; not for real-time motion feedback.

3.2 Recognition and Outputs



3.3 Operating Conditions and Limits

Pallet material



Reflective surfaces are unsuitable.
Assess black or low-reflectance pallets.

Pallet structure and visibility



Show the full entry face, legs and beams.
Occlusion and missing parts affect positioning.
Thin-legged, deformed or self-built pallets
require assessment before use.

2D model suitability



Train with RGB images of site pallets.
Coverage depends on training and validation.

Entry-face angle



Within 15° relative to the camera.
Angles over 15° are filtered.

3D recognition height



Camera-to-floor height must match calibration.
A height change requires ground recalibration.

3D roll tilt

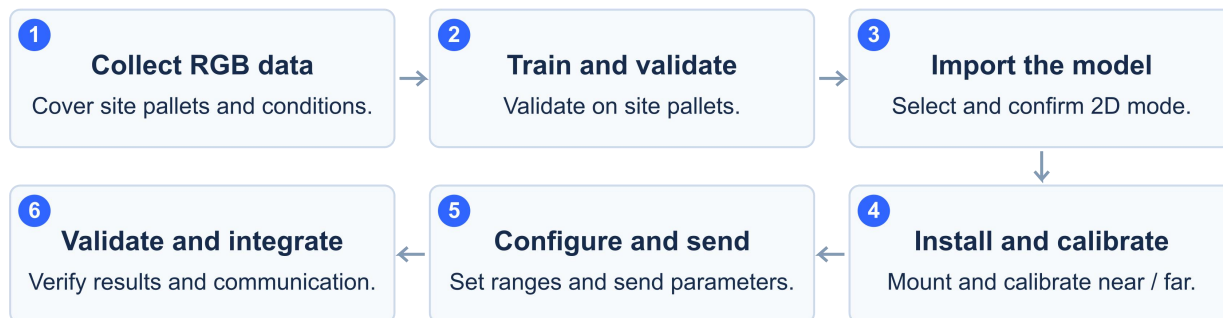


Roll tilt may prevent recognition.
Roll output does not mean all tilts are supported.

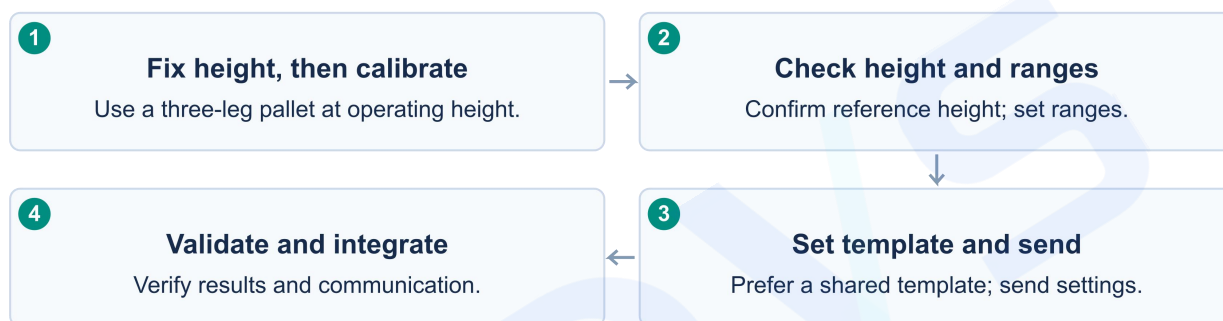
4 Deployment and Configuration

4.1 Deployment Workflow

2D starts with data collection



3D starts with calibration



Computing configuration



5 Validation and Acceptance

Agree on project criteria for positioning error, repeatability, recognition performance and detection cycle time before acceptance. Verify them using the workflow and checks below.



Acceptance item	Coverage
Recognition	Actual pallet types/materials, near/far distances, angles and occlusion; check missed and false detections.
Positioning error	Compare X, Y, Z, Roll and Yaw against references; check repeatability.
Response time	Plan complete detection cycles by mode and platform. Validate result updates and vehicle wait times/timeouts; allow site-specific margin.
Functions / communication	Validate multiple-pallet detection in the mode used. Test load status only in 3D with load detection enabled. Check target identity, status, units and result updates at the receiver.
Recovery	Test no pallet, invalid results, disconnection and recovery; reject invalid or stale data.

6 Frequently Asked Questions

Question	Answer
Does 2D mean planar output only?	No. "2D" refers to image-model recognition. Depth point clouds still provide 3D position and usable orientation.
Can one 2D model cover all pallets?	No. Train on site pallets and validate actual materials, distances, angles and lighting.
Does 3D automatically replace failed 2D?	No. Select and configure 3D separately, then calibrate and validate it before use.
Which camera reaches farther?	M4 Mega: up to 3.0 m. S11 + RK3588: up to 1.8 m. The same limits apply to 2D and pure 3D; validate the usable range with site pallets and models.
Are black and reflective pallets supported?	Reflective surfaces are unsuitable. Evaluate black pallets with the selected camera and actual site samples; appearance alone does not establish suitability.
How are 2-, 3- and 4-leg pallets supported?	Both modes support them. Use a suitable model for 2D or the matching leg-count template for 3D. Pure 3D near/far calibration still requires a three-leg pallet.
What do Closed and Open pallet mean?	Here, Closed has a continuous bottom board joining the legs below the fork openings (With Lower Beam); Open has none (No Lower Beam). This 2D classification applies only to the entry face visible to the camera, not the whole pallet base.
Is real-time motion feedback supported?	No. Both 2D and 3D have latency. Set wait times and timeouts from each mode's maximum total response time. Reject stale results.
Does load detection measure weight?	No. Only 3D, with load detection enabled, reports Empty or Loaded. It does not replace weighing or safety checks. 2D does not support load detection.

Revision History and Support

Version	Date	Description
V0.1	2026-09-28	Measured performance, camera specifications, applications and operating limits.

Technical Support

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GitHub: <https://github.com/Lanxin-MRDVS>

LinkedIn: <https://www.linkedin.com/company/mrdvs>



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